

Developing an infrastructure to support communities of learning

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Through the National Grid for Learning (NGfL) the government is calling for new thinking about the potential of current educational institutions to support lifelong learning. One aspect of the reform is the creation and connection of a learning 'community'. However, knowledge of best practice for implementing reform will only come about through understanding the processes that occur in communities of learning, and, from this, developing an infrastructure to support these processes.

The BT Laboratories (BTL) research project, 'HomeLearn', examined the impact of an on-line educational service on a community of parents, children and teachers. The creation of a home/school network was the first step in involving the wider community in the learning process. The three-month trial demonstrated that children were enthusiastic and creative in their use of communications technologies in the home. They were extremely confident communicating with adults over a network; they learnt to organise meetings, to manage their own time and to resolve conflicts. Some of the families integrated the information communications technologies into their everyday lives. They gradually built up knowledge of the technology through experience and through learning from other members of the family.

From a service provider perspective we learnt that the critical concept in the development of a learning infrastructure is the support of 'relationships', a concept which entails more than simply providing a communications link or displaying static information. The widespread availability of new technologies means that people have an ever-increasing choice about how they want to communicate and with whom. An infrastructure to support learning must provide choice, be dynamic and be personalised. It is only by passing control of the system over to the users that a truly living system will exist.

1. Introduction

Papert [1] describes the computer age as being as big an influence on education and learning as the introduction of reading and writing. Indeed few people involved in educational applications of information communications technology (ICT) would deny the potential it offers for creating a very different type of learning environment from the ones that exist today.

Government, educationalists and industry must work together in order for ICT to be implemented effectively in learning environments. Currently, most research projects around the world are aimed at understanding how ICT can be applied in the classroom. Few projects examine how learning contexts outside the classroom can also benefit from ICT. The BT HomeLearn trial aimed to address the issue by investigating the 'Home' as a future social learning environment. This paper describes the background to the project in terms of current thinking and in the context of the Government's National Grid for Learning (NGfL) initiative [2]. The HomeLearn trial is described as a case study to illustrate how online technologies in the home could be effectively utilised to create a stronger link between homes and school and as a learning tool for families. The creation

of this home/school network is the first step in involving the wider community in the learning process.

1.1 The Government's perspective

The NGfL is a government initiative aimed at harnessing new technologies in education and increasing Britain's international competitiveness. The NGfL consultation paper states that:

'... by 2002 the UK should be a centre for excellence in the development of networked software content for education and lifelong learning, and a world leader in the export of learning services.'

Furthermore, the grid will:

'... enable parents to access general school information and to send messages to the school.'

and the second stage of the grid would:

'... maximise the use of home and school links.'

Therefore, in addition to training teachers to make effective use of ICT in the classroom, a key vision of the NGfL is to involve local communities and families in the learning process by fully utilising the grid.

Recent government reports further emphasise the importance of home study. They recommend that children in junior school should spend 40 to 60 minutes per evening on homework. The findings from a recent US study by Cooper et al [3] show that pupils spent less than half the recommended time on homework per evening (22 minutes) and that this was less than either teachers or parents estimated. Perhaps not surprisingly, Cooper et al also found that teachers' and parents' attitudes towards homework were more positive than those of the pupils.

1.2 *Research approach at BT*

The researchers working in the education and training programme at BT Laboratories (BTL) use 'action research' to explore technology-rich environments. Action research uses practical interventions outside the laboratory, and requires a 'test-bed' approach. The test-bed is used as a model of the future, allowing researchers to observe real teachers and learners as they make innovative use of the technology in their normal working day. As a result, researchers are benefiting from unique insights and are developing a greater understanding of what the introduction of ICT will mean in terms of the day-to-day lives of teachers and learners. BT can use this knowledge to work side by side with the education community to extend the many existing powerful teaching and learning methods, and build the infrastructure for an integrated set of on-line tools.

1.3 *Learning as a social activity*

To produce good tools to support learning, one must first understand the nature of the learning process. Recent research has shown that learning is more than just the processing and retention of information [4–6]. Deep learning requires understanding. Cognitive psychology research shows that knowledge must be integrated as into the framework of existing knowledge — a process that can require considerable cognitive effort from the learner [7, 8].

The constructivist approach to learning assumes that people 'learn by doing' and by iteratively constructing and restructuring their own framework of understanding based on their experiences [9, 10]. According to this view, the importance of learning within a social context is emphasised. Pioneer educational psychologists such as Piaget [11] and Vygotsky [12] proposed that individuals actively construct knowledge by striving to solve real problems within a dynamic social environment. Social constructivism takes the Piagetian position further by suggesting that learning about the social and physical worlds are not separate processes. Social constructivists

suggest that children learn about the physical world through the social world [5].

A consequence of the social constructivist theory is to move the emphasis away from the individual learner to the learners within the social group. Teachers, parents and peers become not only important sources of information, but also significant co-learners and teachers within a social setting.

On-line services can build on the social constructs of learning by providing learners with tools for communication. A key driver within the BT research to date has been the fostering of a social learning context through collaborative on-line projects. Outcomes from BT learning projects include improvements in children's self-confidence and self-esteem, and increased intrinsic motivation [13–15]. It is these 'soft' issues that produce increased learning, rather than any direct impact on specific cognitive processes.

The emergence of the Internet is one of the most exciting technological developments of the last 20 years. To date, the Internet has been described primarily as an information/entertainment source. More recently, research has been emphasising the importance of on-line communications tools and the scope for collaborative working and learning that they offer. Research both within and outside the field of education is highlighting common patterns of Internet use. The HomeNet trial [16] examined use of the Internet within a sample of 110 demographically diverse households. The research suggests that interpersonal communication does, and will continue to, dominate residential use of the Internet. Richards and Keppell [17] suggest that the benefits e-mail brings to education as a feedback medium makes it 'an addictive communications tool'. They describe how, with experience, students optimise the use of e-mail, for example by including abbreviations or emotional icons (emoticons), such as :-) representing a 'smiley' face.

Asynchronous communications tools (e.g. e-mail or discussion groups) have their limitations in supporting learning conversations or dialogues. Fowler and Mayes [18] argue that real-time communications are critical for supporting immediacy of response in collaborative learning situations. In particular, they stress the importance of real-time audio to support co-operative work (e.g. jointly working on a shared document). They argue that learning environments are impoverished by the current quality of real-time communications available over the Internet.

To date the strong emphasis has been on understanding how people learn in the classroom. This is not the only learning environment. The recent drive from government, teachers, parents and researchers has been to look outside the classroom by exploring learning possibilities within the home and community.

1.4 On-line learning at home

Some attempts have been made to make the link between classroom and home learning stronger. In the Apple Classrooms of Tomorrow project (ACOT), teachers offered pupils non-networked computers for homework. Even the students who were normally very enthusiastic were unwilling to accept the teacher's offer [19]. One hypothesis that could be drawn from the drop in enthusiasm is that the social context of classroom work had a critical influence on pupils' levels of enthusiasm. Therefore, in order to maintain high levels of enthusiasm, on-line education services need to support a social learning context in the home by supporting parental involvement and enabling peers to interact across the network.

The Canadian Technopals project [20] reported successes in using on-line technologies in the home. Students were given curriculum-based projects and were encouraged to work together on questions and use e-mail as a common means for discussing project work. The results indicate that the presence of a parent or other person working with the student increased the length of time spent per session on the computer (from 29 minutes to 45 minutes). Subjective findings suggested that the programme also had a positive effect on parental interaction with the children, and that children had a better attitude to both learning and school after the programme.

The remainder of this paper presents the findings from the HomeLearn field trial which was set up to determine the impact of a managed ICT educational service among groups and individuals within a local community in Suffolk. The findings are from an evaluation of the first three and a half months of the trial, from January to April 1998.

2. HomeLearn — a case study

Previous research informs us that the key capability to be supported in an educational service is that of developing a social learning environment. Parents, siblings, peers, experts and teachers can all be critical contributors in the social context. For example, peer tutoring can increase levels of confidence and motivation; an audience or customer for work can increase its quality, and parents and teachers can provide essential emotional and technical support.

The overall aim of the HomeLearn study was to develop a better understanding of how the NGfL could support home/school links. This was viewed as the first step to involving the wider local community in the learning process. This was to be achieved through the following objectives:

- to examine communications behaviour within the home — specifically child/parent,

- to examine communications behaviour across the home/school link — specifically child/child, child/teacher, parent/school and parent/parent,
- to examine the educational/pedagogical implications of bridging a home/school network — specifically educational benefits for the children, problems for teachers in supporting learning over network, and parents' involvement with homework,
- to apply this knowledge in beginning to define a comprehensive managed service to bridge the home/school link

2.1 The triallists

The triallists consisted of eight families with pupils in years 5 and 6 (9 to 11 year olds) and two teachers from Wickham Market Primary School, near Ipswich, Suffolk. This school has been involved in a BT Education Research project since 1994. One of the teachers was a major contributor to BT's schools test-bed and was a regular user of ICT in the classroom. Half of the children were in year 5 and half in year 6. Seven of the children were assessed by the teachers to be between average to above average ability and one was of below average ability.

The triallists all lived within the same area and had pre-existing relationships with at least two other triallists. All the families belonged to a range of community societies and organisations, such as sports clubs, and the Village Hall committee.

2.2 The learning technology

Each home was provided with an additional telephone line, a Macintosh¹ personal computer, printer, digital camera and access to the Internet.

The learning platform was based around the Internet. The system was developed at BTL and was called Real-time Interactive Social Environments (RISE). RISE combines the functionality of advanced voice telephony with dynamic Web browsing, allowing groups of users to talk to each other, record and playback conversations, and view Web pages together during a session. Users are able to see who else is currently on-line. The RISE system includes static Web pages and dynamic information which is customised for each user. RISE also incorporates a proprietary

¹ The computers were donated to the project by The Apple Corporation.

conferencing system called FirstClass, which supports GroupWare, e-mail and discussion groups.

RISE is generic and can be easily customised to accommodate new services. Previous services include one for teaching and learning English as a foreign language. This was evaluated in a project called Merlin [21], which was run in conjunction with Hull University. Another service based on RISE, CollegeComplete, offered an on-line NVQ course for a group of 16 Further Education Colleges using their own intranet [22].

The RISE system in this trial was known to the trialists as the 'HomeLearn' system. The system structure and interface were redesigned to reflect the differing needs and roles of teachers, parents and children as users of the system (see Fig 1).

2.3 Primary functions and features

The main areas of functionality offered by the RISE-HomeLearn system were:

- personalised homework and project work folders for each child, with access restricted to the parents, the child and their teacher;
- personal e-mail,

- e-mail discussion group areas for selected user groups ('Kids Club' for children, 'Parents Room' for Parents, 'HomeLearn Development' for researchers and teachers),
- homework schedule, containing details of the homework, space for comments from the teacher, and icons for parents and teachers to indicate that homework had been seen,
- a 'meeting place' (Fig 2) where it was possible to set up audio conferences with up to 32 people,
- personal recording tool,
- access to short-term, medium-term and long-term lesson plans,
- link to the school's HomePage.

By providing content, communications and information tools, HomeLearn supported the learning process. However, unlike many other educational services, it did not provide any 'pre-packaged' content. The teachers, children, parents and BT researchers provided the content on the system. They were able to update and add to this content at any stage. For example, when appropriate, the teachers could direct the children to a particular Web page and gradually build up links to content as the learning process advanced.

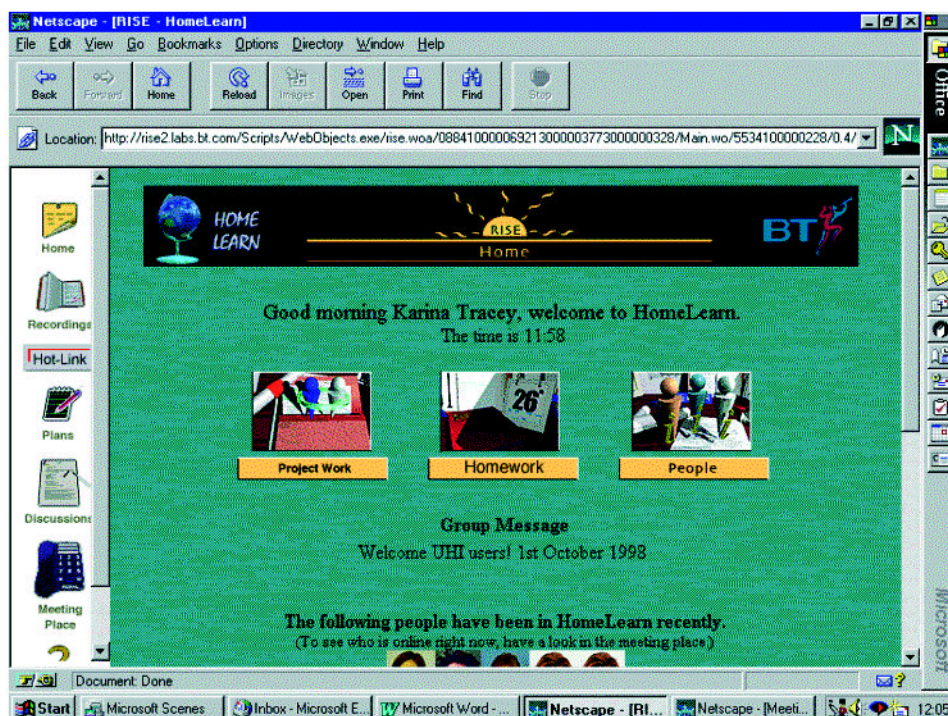


Fig 1 The RISE-HomeLearn Home Page



Fig 2 The RISE Meeting Place.

Trial progress

The equipment was distributed to the trialists in December 1997. Each home had two training sessions with BT. Near the beginning, middle and end of the trial, meetings were held to discuss any issues or problems and to discuss ideas for future activities.

The evaluation of the trial used a multi-method approach. The parents attended a focus group in December 1997. All family members completed pre- and post-trial questionnaires and were interviewed in their homes twice during the trial. The teachers regularly met with researchers, both face to face and on-line, and valuable feedback was obtained through these meetings. They were formally interviewed once at the end of the trial.

In addition, the HomeLearn system recorded all log-ins, logouts, dates and times of use, participation in conferences, participants initiating conferences and incidence of audio recordings.

At the start of the trial the teachers posted two projects on to RISE, one about meteorology and the other on a book that was being studied in class (The Herring Girls [23]). Throughout the trial, the teachers set the children tasks relating to these projects. These tasks were posted into the project folders. For example, the following is a message posted by a teacher in the 'Weather Project' folder:

'Here are the people and schools I would like to try to strike up a relationship with over the weather project. Could you please introduce yourself and send them your RISE e-mail address and ask them to send you their January data....'

Homework was posted on the homework schedule. Once completed the children put the work into their individual homework folders for the teacher to review. Figure 3 is an example of homework posted in the homework schedule.

Three 'Experts On-line' audio-conferences were organised by the BT team and the teachers. The conferences were:

- a group interview with the author of a book the children were studying at school,
- a group interview with a Meteorologist (from the British Meteorology Office),
- a conference entitled 'Technology in the Future', which was chaired by a researcher from BTL.

Prior to the conferences, the teachers set the children the task of preparing questions. During the book author and meteorologist conferences a teacher facilitated the sessions. Below is an extract from the interview with the meteorologist:



Fig 3 The Homework Schedule.

John²: I've just got a question here ...
 Teacher: Who is that, John?
 John: Yes John, um.
 Teacher: Alright.
 John: What is precipitation?
 Meteorologist: Precipitation? Anything, which falls from the skies really, so it can encompass rain, snow, drizzle, hail ...
 John: Frogs ...
 Meteorologist: It could actually include frogs and there were some of those not so long ago falling out of the sky.
 John: Cats and dogs.
 Meteorologist: Yes, cats and dogs, yes. (laughs)

The audio conferences were recorded and later used as a source from which homework could be set. Below is an example of homework that was set using the book author interview:

'Listen to 'Herring Girl' recording with another Home Learn group member and agree the main points from the conference. How do these compare with other people in your group? Have you identified the main points? Bring a copy of your final version to school to share with everybody in year 5/6.'

The conference recordings could also potentially be used as a source of material for new learners in future years.

² The real name of the child has been changed.

These new learners could learn vicariously by listening to the experiences of the children (see McKendree et al [24] for more details about vicarious learning).

2.4 Background, expectations and motivations

Prior to the trial all the families owned a computer, six already had access to games, and two to the Internet and e-mail. Mothers were the main telephone users, making 58.2% of the calls. The children made just 4.1% of the calls.

The families were motivated to participate in the trial in order to gain experience using ICT and to use the Internet. They expected a considerable amount of computer time to be spent on leisure, either playing games or browsing the Internet. The parents hoped that they would be able to use the computer to communicate with other parents, contact the school and help their children complete homework when the trial started. In particular, they felt an electronic 'Homework Schedule' could provide them with a tangible connection to their child's classroom.

'What mainly happens is that I say to [year 5/6 child] 'Have you got any homework tonight?' and he says 'No,' and then I look in his bag and he has got some after all. So this could be really useful to check if he has homework and make sure he has taken it down correctly.' (Father at pre-trial focus group)

Some concern was expressed about possible addiction to the Internet, and about monitoring the nature of the material children access on the Internet.

3. Findings

3.1 General usage of the ICT

All the triallists felt they gained some valuable experience and knowledge in using ICT. Families reported that the primary use of ICT during the trial was for leisure purposes. The main leisure activity for parents was browsing the Internet. All the children reported browsing the Internet for pleasure and a majority of the children played computer games.

Table 1 shows the total number of log-ons for each group of users, per month and over the whole trial. The numbers in brackets are the mean per individual within the relevant user group (i.e. mothers, teachers, etc)

Table 1 also shows that, per individual, the teachers were the highest users of the HomeLearn system. The children were the highest users from the families and there were slightly more log-ons for mothers than there were for fathers.

The data also shows a particularly striking percentage drop in log-ons for parents from January to February with 46% drop for mothers and 36% drop for fathers. Primary reasons given by parents for accessing the system were to check for e-mails and to look at the homework schedule. The parents attributed the drop in use to a low number of homework schedule updates, to receiving few e-mails and to finding the e-mail inconvenient to access as it was embedded several layers into the HomeLearn system. Generally one would expect a usage drop as the novelty wore off, and people spend less time browsing the system in order 'to see what is there' and began to log-on primarily when there was something specific they wanted to access.

Year 5/6 children mainly logged on to check for and send e-mails, to send homework to the teachers and to participate in audio conferences.

Triallists spent an estimated three quarters of their time using ICT alone, and one quarter with others.

3.2 Individual and family differences

There were large variations in frequency of usage within the group of year 5/6 children and within the groups of parents.

- the year 5/6 children logged on between 22 and 196 times over the course of the trial,
- mothers logged on between 0 and 37 times during the trial,
- fathers logged on between 1 and 45 times.

Analysis of log-ons across the families showed a strong positive correlation between children's use and their parents use of the system. This suggests that family influence affected individuals' frequency of access to the system. Kraut et al [16] also suggest that social influences affect Internet usage in this way.

Interview data suggests that all but one of the families adopted a 'culture' of ICT use as part of their evening and weekend routines. These families could describe the typical evening routine and the times of the day when each member of the family would normally be using the computer. They were generally aware of who was using the computer and what they were using it for. This awareness was reported even when the computer was situated in a child's bedroom.

Use of the computer by 'ICT-adopting' families had a strong educational emphasis. The computer tended to be viewed as belonging to the children, and they were given access priority. Regardless of application, any activity using the computer was considered to be more educationally beneficial than passively watching the television.

Table 1 Numbers of system log-ons during the HomeLearn trial.

	January (total of 26 days)	February (total of 28 days)	March (total of 31 days)	April (total of 6 days)	Combined total (91 days)
Teachers	78 (39)	77 (38.5)	97 (48.5)	10 (5)	262 (131)
Yr. 5/6 Child	260 (32.5)	236 (29.5)	185 (23.1)	13 (1.6)	694 (85.1)
Mothers	75 (9.4)	40 (5)	46 (5.75)	6 (0.8)	167 (20.1)
Fathers	67 (8.4)	43 (5.4)	33 (4.1)	6 (0.8)	149 (18.6)
Siblings	31 (2.3)	21 (1.75)	20 (1.7)	1 (0.1)	73 (6.1)
Total	511 (13.4)	417 (11.0)	381 (10.0)	36 (0.9)	1345 (35.4)

'... it would be much better than watching Eastenders ... because there is some input from them if they are using the computer. With television you just receive it, the computer involves a little bit of activity from them.' (Father in an interview)

3.3 Roles in the home

All of the families had a 'family expert', usually the year 5/6 child or the father who was asked for help every time a technical or operating problem was encountered. In general, the families seemed to provide technical support for one another, particularly during the early ICT adoption phase. The following extract from a dialogue in one of the family interviews illustrates this point:

Father: I think at first I was called up there a lot.
Mother: Yes, but it is definitely not as frightening now.
Father: I know...besides [Yr 5/6 child] is very good, he often helps his mum and [older brother].

The main source of educational support for the children came from the mothers. As opposed to being the ICT experts, they reported learning and using the ICT alongside the children. In three of the cases the parents reported that the children spent longer per session when using ICT with their parents or a friend. They attributed this to the child having a low attention span, or a lack of confidence in using ICT. In these cases the desire for social, face-to-face support was consistently high.

3.4 Communication across the network

Parent to parent

In comparing contacts before and during the trial, the number of telephone calls and face-to-face meetings among the parents remained constant after the introduction of e-mail and audio-conferencing into the homes. Despite knowing at least two other trialists, the parents did not participate in audio-conferences with each other, and very few e-mails were exchanged. Mothers appeared to be the primary maintainers of relationships outside the family and among them there was a strong resistance to replace telephone or face-to-face contact by e-mail. They felt that some emotional context would be lost in e-mail. Also, the parents were not confident that the other parents were checking their e-mail on a regular basis, so they used their usual communication methods to contact the other families instead.

Home to school

Teachers often came on-line in the evenings simply to display their availability for contact. Although the families rarely contacted teachers, the teachers sometimes contacted

the children to check on their progress and offer assistance. Being obviously visible on-line through the photographs in the meeting place was felt to be essential for the teachers in their support role.

Parents reported that many of the interactions they had with the school were related to 'sensitive' issues and therefore they preferred to speak to the teacher face to face.

'I was going to e-mail one of the teachers about [yr 5/6 child's] homework, but I feel some things are rather more delicate and better handled face-to-face.' (Mother in interview)

E-mail was used by the parents in businesslike interactions with the school. For example, when an unexpected event occurred, such as an injury, parents needed to liaise with the school. However, during holidays when the school was closed, this would normally have been difficult.

'...when [sibling] broke her wrist it was half term, nobody was at school and we really wanted to know if he could go back to school ... so we e-mailed the teacher as it was her holiday so picking up the telephone wouldn't have been right ... this way she can look at it when she wants to.' (Mother in interview)

By e-mailing the teacher directly instead of phoning the school office to leave a message, the link between the home and the school was personal, without being intrusive.

Contrary to expectations prior to the trial, e-mailing teachers in the evening was not felt to be intrusive, as it was more like leaving a message than actual one-to-one communication.

Child to child

The children frequently made use of e-mail and audio-conferencing during their evening routines. They appeared to refine their use of the tools over time. For example they learnt that conferences without fixed agendas and set times were often unsuccessful. The 'Experts On-line' conferences were highlights of the trial, not only because they provided an authentic learning experience, but because the teachers set the children specific pre-conference tasks and the teachers moderated the conference, ensuring that the children remained focused on the task.

The children varied in the importance they placed on on-line activities in their routines. One child intuitively learnt to arrange his on-line sessions so they would coincide with his friends' sessions.

'When I get home from school because I know nobody else will be on the computer I watch TV for a bit, then when I think everyone else will be home I go up and

go on the computer.’

The mother of this child suggested that he/she used it as a substitute for face-to-face peer contact in the evenings.

‘The communication aspect of it is very important to [child] as there are no other young children around here who are [child]’s age. For his brother it wouldn’t be so important as he has a group of friends that live next door and round the corner.’ (Mother in interview)

E-mail was used regularly among the children, and between teacher and child. E-mails tended to be short and related to specific subjects.

Personal e-mail was used:

- by the children to send homework to the teacher,
- by the teachers to give instructions or hints about homework,
- by one child to send his ongoing project work back and forth between home and school,
- between pairs of close friends, on a regular basis and to continue on-line themes of conversations: ‘We have an ongoing conversation about things, some of it is social and some of it is work — I would e-mail him and he would reply every day,’
- between the children to continue ongoing face-to-face themes of conversations: ‘... if I want to send him something right away I usually send him an e-mail, he usually checks it in the morning before school and then when we get to school we can continue talking about it.’ (Yr 5/6 child)

The overall amount of communication among the children increased. Previously the children were infrequent telephone users, making calls at the most once per week. With the introduction of e-mail and conferencing, the children became much more active communicators in the evening, conferencing on average three times per week and sending up to three e-mails per day. Most of the e-mails and conferences were oriented around general chat. However, some dialogues had a mix of school-related and personal topics.

The children made new friends and strengthened existing friendships. Six out of eight children reported that they had got to know some of the other trialists better. Relationships among the children were built up over time and through a combination of face-to-face contact during school and e-mail and conferencing in the evenings.

3.5 Educational impact

Comparisons of estimated time spent on homework before and during the trial showed an increase from a mean of 95 minutes per week to a mean of 135 minutes per week. The children estimated that when using the computer alone they spent 35% of their time on education-related activities as opposed to 47% when using it with others.

Overall there was an increase in the mean length of time parents spent helping their children with schoolwork, from 21 minutes per week prior to the trial to 50 minutes per week during the trial. Typical parent/child activities included exploring the Internet as a resource for homework, and using software packages such as Excel.

Parents, teachers and the children themselves reported positive outcomes as a result of the children using ICT. Through using e-mail and audio-conferencing to communicate with others, the children refined their verbal and written communication skills and learned about the importance of presentation and organisation of content.

‘Looking at the work that [year 5/6 child] produces on paper and looking at the work he does on the computer, it seems he can organise a story better on the computer It is probably because he can read it back and shuffle his ideas around.’ (Mother in interview)

The primary overall benefit for the children was reported to be an increase in confidence. As the following quote from a parent shows, the increased anonymity of on-line communications can have beneficial effects.

‘I think she would approach adults more readily through the computer...I have noticed that she communicates with [teacher] and [BT researcher] about lots of things and I don’t think she would have done that face to face...she is more equal and is quite casual about it.’ (Father in interview)

The children started to develop some core lifelong skills. For example, during the trial an argument arose in the kids club conference area about whether boys were better than girls. The children had to struggle to keep this conflict under control and learn to become more sensitive to other’s reactions. The children recognised that although the anonymity of e-mail increased their confidence, it meant they could be desensitised to others’ reactions over e-mail. In the same way that adults who use Internet chat rooms can take on a whole different identity when on-line, the children were sometimes very different when interacting over e-mail as opposed to face to face. The following extract taken from an e-mail in the ‘kids club’ conference illustrates the point:

‘I’ve been noticing the change in the kids club with arguing because [female child] has been saying stuff

like “girls rule” and “boys don’t know how to have fun” ... and at school she’s all nice and doesn’t say that girls rule ...’

The children who most frequently sent e-mail and joined in audio conferences were reported to gain the most from the experience. In particular, the teachers and parents reported that the more the children experimented and innovated in their use of ICT, the greater the rewards. One of the children, who was described by the teachers as being very able, only accessed the ICT infrequently, and as a result the benefits yielded for this child were low. This child did not experiment with the ICT, and according to the teachers was not interested in joining in with the e-mail and audio-conference discussions.

3.6 Problems for the families

The commitment required from the parents was higher than expected. It was hoped that parents would use the system not only with their children but also to communicate with other parents and the teachers. In the latter case, the only time most parents had available to use the computer was in the evening after the children had gone to bed. So, if the computer was situated in the child’s bedroom the parents were often unable to use it.

An important factor in any small-scale project is reaching a critical mass of users. It was clear that since few of the parents were close friends before the trial, relationships within the group were hard to form within a time span of three months. This factor, in combination with their low level of direct involvement in the projects, a lack of received e-mail, and the infrequent updating of the Homework Schedule caused a decrease in parents’ use over time.

‘You start off going in every day to check and it is the same every day, so you start looking every other day and it still hasn’t changed so you end up leaving it, and then even a week later it still hasn’t changed.’ (Mother in interview).

In these circumstances, interactions within the community may need a formal ‘push’ to overcome the initial inertia. Perhaps involving the parents more explicitly in the early homework projects, or setting up a series of managed sessions between teachers and parents could have helped to seed and encourage further interactions.

3.7 Impact on teachers

Table 1 showed that, per individual, the teachers logged on the greatest number of times during the trial. Involvement of only eight children from a year group of 60-80 meant that the HomeLearn approach was not fully integrated into the classroom. One consequence was that the

teachers sometimes failed to update the electronic Homework Schedule, finding it easier to give homework using the blackboard. Additionally no short-term plans were ever used, as their inclusion on the system proved to be too time-consuming for busy teachers. However, the teachers were confident that if the system were used for all the children in the class, it would provide a powerful and beneficial learning environment.

Since only two of the teachers had access to the system, problems arose when they needed to co-ordinate activities such as homework planning and project work. The teachers considered it crucial that any HomeLearn activities were integrated into the planning.

The teachers appeared to have some difficulty in incorporating the new technology into their teaching approaches, and also in using its flexibility to provide the parents with a broader view of their children’s education. It seemed they needed to make a conceptual shift from focusing on the child/teacher relationship to focusing more on the family/school link. Teachers needed support to open the classroom to other influences and learners, such as the parents and siblings.

3.8 Implications

Communications behaviour across the school/home link

Many factors will affect the nature, frequency and intensity of communications across the school/home link [18]. For instance, the relationship between individuals may require different levels of formality. If the teacher and parent know each other well, they will be more prepared to use lower quality communication, because of the reduced risk of misunderstandings. The content of the communication may also be critical. For example, parents were reluctant to e-mail teachers about sensitive matters relating to their children’s academic and social well-being, but were more than happy to use e-mail as a mechanism to interact on more business-like matters, such as reporting illness or injury. Therefore, when developing services, thought should be given to how a range of technologies can be integrated to offer different styles or ways of interacting depending on, for example, who you are talking to, what time of day it is or what you want to communicate about.

It is also important to understand the interactions between different types of communications technologies and face-to-face meetings. For example, there were indications that the children e-mailed others more regularly if the relationship was based on face-to-face contact. Interactions seemed to snowball when children had the opportunity to continue e-mail and conferencing discussions through face-to-face interactions at school. Replacing face-to-face communications with on-line communications may not be the best approach. The real

benefits may be found through increasing the overall level of interaction by enhancing existing face-to-face interactions through e-mail and audio-conferencing. Designers of ICT systems need to be much more aware that they are creating socio-technical systems, and that to be successful, these have to be based on a rich understanding of the life-styles of the various stakeholders.

Communications in the home

The social environment in the home was a stronger indicator of level of ICT adoption than peer influences. Therefore, understanding the relationships within these micro-communities is essential for understanding how the home/school link could be strengthened.

Within the home 'organisation', developing a culture of ICT adoption involved gradually refining use, learning from past personal experiences and, critically, learning from other members of the family. The families moved through distinct stages of ICT introduction — adoption by individuals, acceptance as part of the family routine, and finally innovation and experimentation using ICT. During uptake of ICT, different members of the family took on roles as technical experts, motivators, coaches and emotional supporters.

Changing roles

Classroom-based ICT projects show that teacher/pupil roles were changed by the introduction of ICT [15, 19]. Similarly, in the home, the roles of the parents and children were seen to change. Frequently, parents were consigned the role of learner, recognising that their children had greater knowledge and skill than themselves. This was a powerful experience for the children. Their self-esteem was raised, and they were enthusiastic tutors, as often they had just learnt the skills themselves and were excited about sharing them with others.

Peer relationships also changed as the children began coaching each other in school and through e-mail in the evenings. The children acted as role models for each other. They learnt from each other, and provided each other with satisfying positive reinforcement. Again, similar to the findings in classroom-based projects, the sense of having an audience or customer for work was a source of powerful intrinsic motivation.

Implications for education

In addition to taking on more responsibility for their learning and showing more enthusiasm, both teachers and parents reported that the children developed some very generic skills. They were more confident about interacting with adults, organising meetings, managing their own time and resolving conflicts. The factors contributing to

increased skill development are still not clear — a greater understanding is needed of the conditions that result in children being more engaged and thus developing skills above and beyond normal expectations. The role of internal factors like emotion, attitude, goal commitment and curiosity clearly need further investigation.

Defining a managed system

The triallists found that HomeLearn was a useful platform for supporting the links between the school and the home. However, the trial provided some food for thought when considering how to support learning communities. Although the triallists did not want to vastly reduce or remove existing ways of communicating, they were willing to try the new technologies; but, like meeting new people, first impressions endure. If sending an e-mail to a friend does not have the desired effect first time round (e.g. the person never read it), the sender will be less likely to repeat the action. Adopting new technology is not an altruistic behaviour — people want to know that there is a definite benefit in spending time grappling with technology, whether it be learning about a particular piece of software, investing in their children's education or maintaining a relationship.

There were obvious benefits for the children in being on-line, and this group gained a great deal from their experiences. However, teachers needed to see clearer and more obvious benefits in return for investing time and effort updating electronic systems (such as a homework schedule) and replying to e-mails from parents. The findings from the trial suggest that a managed system would need to provide teachers with a stronger infrastructure of support and some explicit examples of how a system like HomeLearn could be integrated into day-to-day teaching activities. This could be achieved through samples of completed on-line projects, on-line teacher support groups and service provider help lines.

Despite most of the parents in the trial being very committed to using ICT to help their children with schoolwork, their usage and enthusiasm decreased during the trial. The key to maintaining on-line activity was to give people a strong reason to be on-line in the first instance. Once on-line, they usually spent time using additional applications and spent far longer on-line than originally intended. Therefore a system like HomeLearn would need to provide reasons for parents to be on-line other than involvement in their children's education. Impetus could perhaps be created by providing parents with the opportunity to also learn on-line through the local colleges or by bringing local businesses on-line.

The 'Experts On-line' conferences were very successful. However, they were one-off events that created high overheads in terms of time resources for the teachers

and researchers. Experts who are recruited to help in an ongoing capacity must be part of the community, and should see some reward for their sustained involvement. Respected community members with a drive to be involved in local schools are already physically brought into schools to talk about careers and act as positive role models for the pupils. For example, the 'Neighbourhood Engineers Scheme' has been set up around the country for retired engineers who want to promote engineering as a career to school children. This approach could be augmented to include on-line activities, such as presentations, interviews or mentoring through conferencing and e-mail. Additionally parents and friends from families within the local community could be recruited as 'experts' in evening conference sessions.

4. Conclusions

In this trial, the success of the families' and teachers' adoption of the new technology helps us understand how effectively a HomeLearn-like service could be used to support learning in the future. The following are the key conclusions:

- using ICT at home to communicate with other learners, teachers and adults offers powerful benefits for children in terms of personal and social skills,
- families who adopt a culture of ICT use in their home have an influence on each other's level of ICT use,
- individuals in families will take on different roles in the family when using ICT (e.g. emotional supporter, tutor) and sometimes these will be different from their usual family roles (e.g. a child who tutors their parents in using e-mail),
- the success of the service rests very much in the hands of the teachers,
- all key teachers need to be included in the service to allow co-ordination of activities across classes.

This trial is about understanding how the government's vision [3] of networked learning in the future could be practically implemented. 'Learning' is a fitting context for the trial as during the first 3 months it was not only the children, but also the parents and teachers that were being asked to learn something new — using the HomeLearn system to support their relationships and their children's learning activities. From observations of this trial, and of many more carried out in a real world context, it has been shown that by entertaining new ideas and new ways of doing things individuals open themselves to difficulties, doubts and feelings of incompetence, especially when things are tried for the first time.

Although many reported difficulties, all the triallists felt they were successful in learning something about using ICT

to support learning by the end of the trial. However, even if the triallists were as competent and comfortable operating HomeLearn as the designers of the system themselves, it is unlikely that it would make the actual process of 'learning' within the context of education and schools any easier. Rather the potential for ICT is in its power to open doors to other people — for example other learners who are encountering the same problems, a trusted teacher who can help with useful advice, or an expert working in the field that is being studied. So, by enriching individuals' experiences of learning through carefully designed social learning infrastructures, instead of making learning 'easier', the motivation is there for them to explore, to experiment and to innovate, and ultimately to embrace the challenges that learning presents.

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